

HW2 - ER Diagram

Team 59

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Review and evaluation of the requirements analysis document

Completeness

The requirements document provides a fairly comprehensive overview of the system, covering key entities such as neighborhoods, restaurants, gangs, and law enforcement agencies. It also clearly defines the relationships between these entities, making it easier to conceptualize the interactions within the mini-world. However, some improvements are necessary, especially regarding the specification of attributes and the use of composite attributes where applicable. For example, simple types were used attributes defined as complex, like in the case of *address* which was defined as complex, VARCHAR(300). Finally, the cardinality and (min, max) parameters have not been defined for the 3-degree relationship. We have outlined our proposed modifications below.

Correctness

While the majority of the requirements are correct, certain design choices such as the use of ENUM for attributes like cuisine and type might limit the system's scalability and flexibility. By redefining these as composite attributes, the system can better handle more dynamic data types. Furthermore, the weak entities, such as *Gang_Member*, should not have primary keys independent of their strong entities, as this violates relational database design principles. Also, (min, max) parameters for many-to-many 2-degree relationships have duplicate upper limits.

Ease of Modeling

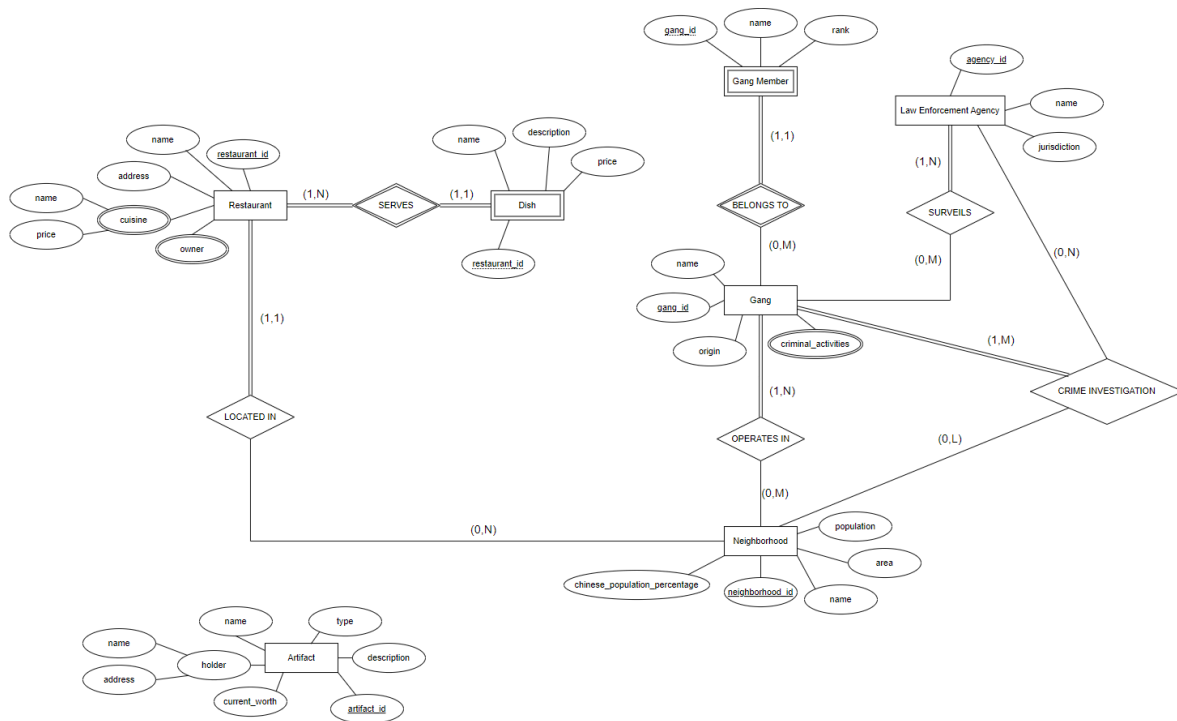
Modeling the mini-world based on the current requirements is relatively straightforward, given that the entities and relationships are well defined. The document provides clear cardinality ratios, constraints, and examples, which would make the implementation easier for developers. However, the proposed modifications—such as refactoring composite attributes and redefining ENUM values—would further enhance the database's ability to model real-world complexity.

While the document serves as a solid foundation for building the database, the suggested modifications would improve both its accuracy and flexibility. These changes will better reflect real-world data relationships and ensure a scalable, well-structured system.

Modifications

1. Standardize Attribute Type for *name*: Change the type of the attribute *name* to simple across all entity types to ensure consistency throughout the schema.
2. Modify Population Attribute: Change the data type of the *chinese_population_percentage* attribute from INT to FLOAT to increase accuracy.
3. Refactor Address Attribute: Convert the *address* attribute from a composite attribute to a simple VARCHAR(300) attribute.
4. Redefine Cuisine Attribute: Replace the *cuisine* attribute, currently defined as an ENUM, with a multi-valued composite attribute that includes name (VARCHAR) and price (FLOAT). This modification allows for more flexibility in representing the cuisine data, such as associating different dishes with varying price points. Similar changes should be applied to other ENUM attributes like *type* (painting, sculpture, etc.), and *rank*, which should be changed to simple VARCHAR(100) and *criminal_activities* which should be changed to multi-valued VARCHAR(100).
5. Remove Primary Keys from Weak Entities: Eliminate the primary keys from weak entities, *Gang_Member* and *Dish*, as their identification should be dependent on the corresponding strong entity, ensuring proper relational design.
6. Increase Description Length: Expand the data type of the *description* attribute to VARCHAR(500) to accommodate longer text entries where necessary.
7. Fix the definition of attribute *holder* in entity *artifact* to a correct composite definition. *Holder* now has sub-attributes *name* and *address*.
8. Add (min, max) parameters (1,M), (1,L), (1,N) to the 3-degree relationship CRIME INVESTIGATION.
9. Fix duplicate upper limits in (min, max) parameters for many-to-many relationships. The 2nd N becomes M.
10. Fix upper limit for *Gang_Member* in relationship BELONGS TO to 1 as it is a many-to-one relationship.

ER Diagram



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[Link](#) to the diagram